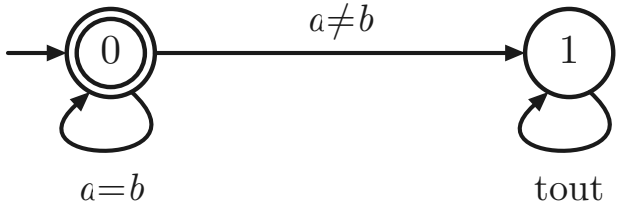
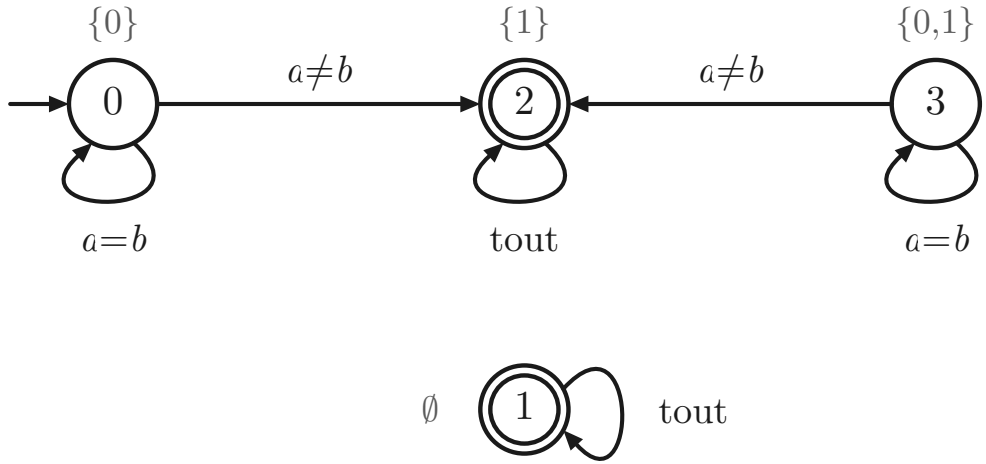
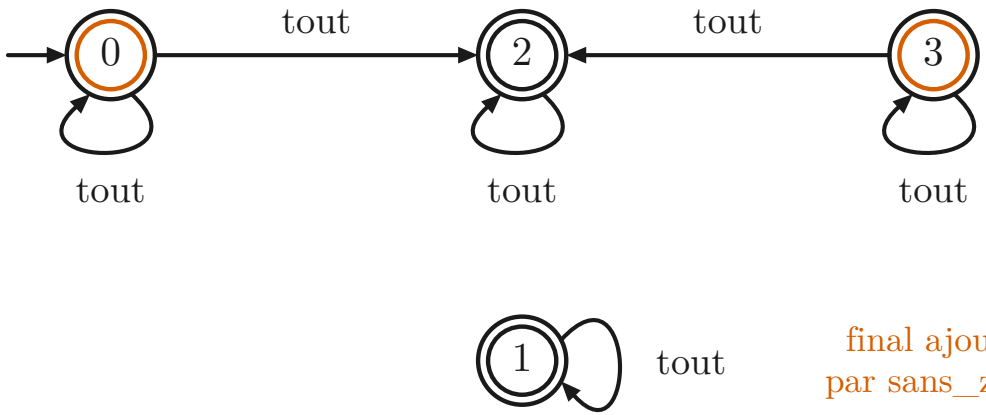
**a** $E$: $x_1 = x_2$



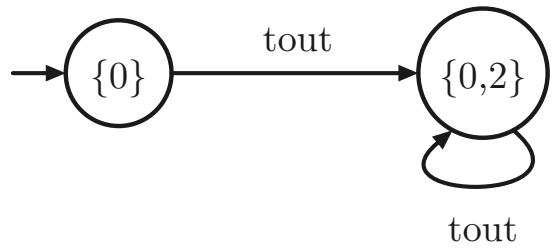

**b** $B$= complémentaire $E$




**c** $C$= existe $B1$, puis $D$= sans_zero $C$




**d** complémentaire $D$




$L_\varphi = \emptyset$